

Aegis Geopolymer Concrete

Cost Analysis per Unit — CSMFAB 107-concrete

Category: Infrastructure — Construction **Unit Basis:** per m³ (batch-delivered, BFRP reinforced) **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
Fly ash (class F)	380 kg/m ³	\$0.12/kg	\$45.60
Metakaolin	85 kg/m ³	\$0.55/kg	\$46.75
GGBS (ground granulated blast slag)	115 kg/m ³	\$0.18/kg	\$20.70
NaOH activator (8M solution)	42 L/m ³	\$0.45/kg	\$18.90
Na2SiO3 solution (Ms=1.8)	95 L/m ³	\$0.55/kg	\$52.25
ZrB2 nano-whisker admixture (5 wt% agg.)	60 kg/m ³	\$350.00/kg	\$21,000.00
YInMn Blue surface pigment (applied separately)	0.5 kg/m ²	\$75.00/kg	\$37.50
BFRP rebar (per m ³ structural equivalent)	25 kg	\$25.00/kg	\$625.00
BOM Subtotal			\$21,846.70

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$21,846.70	\$17,914.29	\$14,200.36
Direct Labor	\$180.00	\$120.00	\$80.00
Overhead (15% of labor)	\$27.00	\$18.00	\$12.00
Quality Control & Testing	\$21.60	\$9.60	\$4.00
Packaging & Logistics	\$655.40	\$546.17	\$436.93

Tooling & Equipment Notes: Batch plant capital: \$450K shared facility. ZrB2 whisker is dominant cost driver.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$23,500	\$37,600 – \$51,700	35-55%
Small Batch (100-500/yr)	\$17,200	\$25,800 – \$34,400	33-50%
Production Volume (5K-50K/yr)	\$13,800	\$19,320 – \$24,840	28-45%

Market Reference: \$250-400/m³ base (without ZrB₂ whiskers: \$500-800/m³); \$4,000-6,000/m³ with EMF-shielding ZrB₂ additive

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB ₂ /Si ₃ N ₄ UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO ₂ route)
MXene Ti ₃ C ₂ Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$3,932 – \$7,646 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

End of Aegis Geopolymer Concrete Cost Analysis | CSMFAB 107-concrete | June 2026